

Guia de instalação da Máquina Virtual e ROS2

A primeira coisa ser feita é escolher a versão a ser baixada, essa informação pode ser consultada em: <docs.org.org> na aba **distributions**

Para atender as necessidades do modelo foi escolhido a versão **Humble Hawksbill (Ubuntu 22.04)**

List of Distributions				
Below is a list of current and historic ROS 2 distributions. Rows in the table marked in green are the currently supported distributions.				
Distro	Release date	Logo	EOL date	ROS Boss
Jazzy Jalisco	May 23, 2024		May 2029	Marco A. Gutiérrez
Iron Irwini	May 23, 2023		December 4, 2024	Yadunund Vijay
Humble Hawksbill	May 23, 2022		May 2027	Audrow Nash
Galactic Geochelone	May 23, 2021		December 9, 2022	Scott Logan
Foxy Fitzroy	June 5, 2020		June 20, 2023	Jacob Perron / Dharin

Acessar o site <releases.ubuntu.com>, para realizar o download

Escolher a versão **22.04.05 Jammy Jellyfish**

ubuntu[®] releases

These releases of Ubuntu

Standard support

LTS Releases	Interim Releases
Ubuntu 24.04.2 LTS (Noble Numbat) ›	Ubuntu 24.10 LTS (Orion) ›
Ubuntu 22.04.5 LTS (Jammy Jellyfish) ›	
Ubuntu 20.04.6 LTS (Focal Fossa) ›	

Select an image

Ubuntu is distributed on three types of images described below.

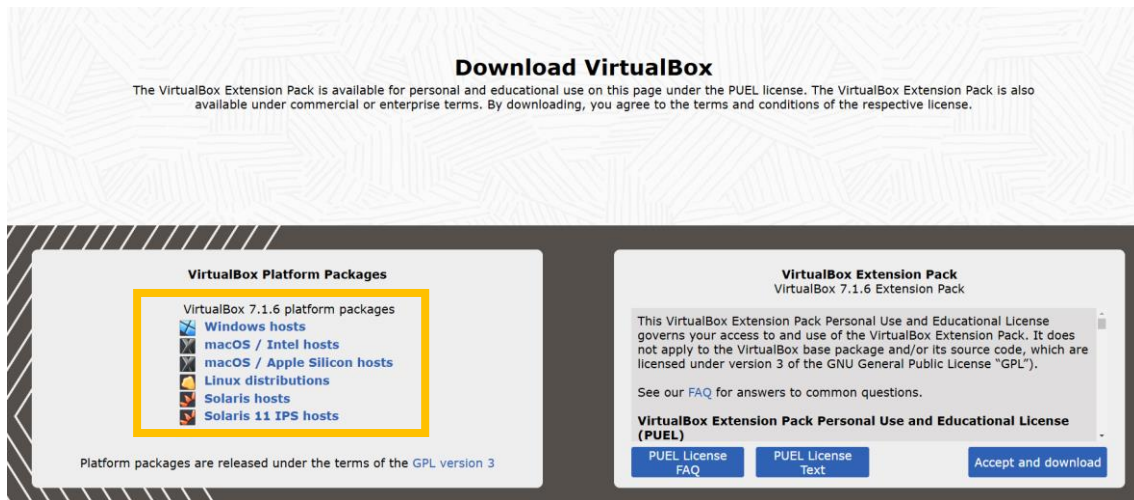
Desktop image

64-bit PC (AMD64) desktop image

Choose this if you have a computer based on the AMD64 or EM64T architecture (e.g., Athlon64, Opteron, EM64T Xeon, Core 2). Choose this if you are at all unsure.

The desktop image allows you to try Ubuntu without changing your computer at all, and at your option to install it permanently later. This type of image is what most people will want to use. You will need at least 1024MiB of RAM to install from this image.

Acessar [virtualbox.org](https://www.virtualbox.org) para baixar a Máquina Virtual, de acordo com o sistema operacional



Executar o arquivo e seguir os passos de instalação

Abrir o arquivo, criar uma nova máquina virtual

New > Type: Linux > version: ubuntu64

Definir o tamanho de memória, isto depende da quantidade de memória RAM do computador (Recomendado 16gb > 6gb; 8gb > 4gb)

Instalar ROS2 - humble

Teste se o computador suporta UTF-8

```
locale # check for UTF-8
```

```
sudo apt update && sudo apt install locales
```

```
sudo locale-gen en_US en_US.UTF-8
```

```
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
```

```
export LANG=en_US.UTF-8
```

```
locale # verify settings
```

Habilitar o Repositório Universal do Ubuntu

```
sudo apt install software-properties-common
```

```
sudo add-apt-repository universe
```

Adicionar a chave GPG do ROS2

```
sudo apt update && sudo apt install curl -y
```

```
sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -  
o /usr/share/keyrings/ros-archive-keyring.gpg
```

Adicionar o repositório a suas listas do comando *source*

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-  
archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu $(. /etc/os-release &&  
echo $UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list  
> /dev/null
```

Atualizar os repositórios do sistema

```
sudo apt update
```

Antes de instalar os pacotes, realizar upgrade do sistema

```
sudo apt upgrade
```

Instalação do desktop: ROS, RViz, demos e tutoriais

```
sudo apt install ros-humble-desktop
```

Instalar bibliotecas base do ROS

```
sudo apt install ros-humble-ros-base
```

Ferramentas de desenvolvimento

```
sudo apt install ros-dev-tools
```

Instalar diferentes bibliotecas e ferramentas do ROS

```
sudo apt install ros-humble-gazebo*
```

```
sudo apt install ros-humble-ros2-control
```

```
sudo apt install ros-humble-ros2-controllers
```

```
sudo apt install ros-humble-xacro
```

```
sudo apt install ros-gz* (novo gazebo - opcional)
```

```
sudo apt install ros-humble-*-ros2-control
```

```
sudo apt install ros-humble-joint-state-publisher-gui
```

```
sudo apt install ros-humble-turtlesim (opcional)
```

```
sudo apt install ros-humble-robot-localization
```

```
sudo apt install ros-humble-joy
```

```
sudo apt install ros-humble-joy-teleop
```

```
sudo apt install ros-humble-tf-transformations
```

```
sudo apt install ros-humble-tf2-tools
```

```
sudo apt install ros-humble-urdf-tutorial
```

```
sudo apt install ros-humble-navigation2
```

```
sudo apt install ros-humble-nav2-bringup
```

Adicionar ao arquivo setup.bash o código abaixo

```
# Replace ".bash" with your shell if you're not using bash
```

```
# Possible values are: setup.bash, setup.sh, setup.zsh
```

```
source /opt/ros/humble/setup.bash
```